NWU®

# Alternatives, Evaluation and the Master Facility Plan.

W Kruger
2025

# Developing alternatives

Design = *CREATIVE* part of the process
(can feel FUZZY)

- May feel like going back and forth (zoom in zoom out) – YES that is OK!
- You can detail only certain areas/aspects
- Your base data/information must be <u>reliable</u>
- Some procedures may help e.g. SLP (Munther)
- There is <u>iteration</u> between steps, even going back to recalculate, measure, fit, move, redesign process

- To test feasibility – SIMULATION models may help

*Ch 6:10 (Tompkins)*

# Developing alternatives

- Involve people
- Think outside of box
- *Brainstorm* and *Bounce* ideas
- Don’t think solution too early
- Don’t fail to see the forest for the trees
- Don’t *judge too quickly*

The DESIGN process ask for a SELECTION to be made, it does not ask for the BEST design! Note that.

Goal therefore is to ‘satisfy’ not ‘optimize’

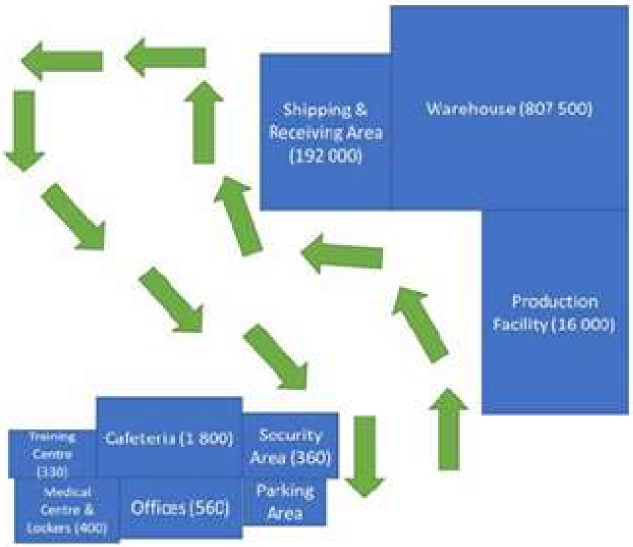
*Ch 6:10 (Tompkins)*

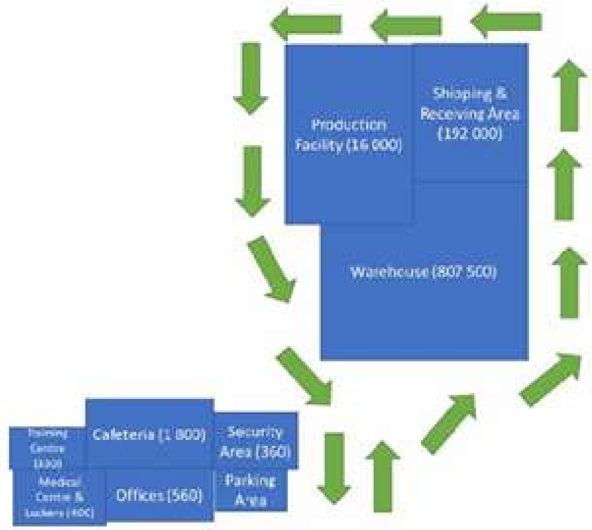
# Developing solutions

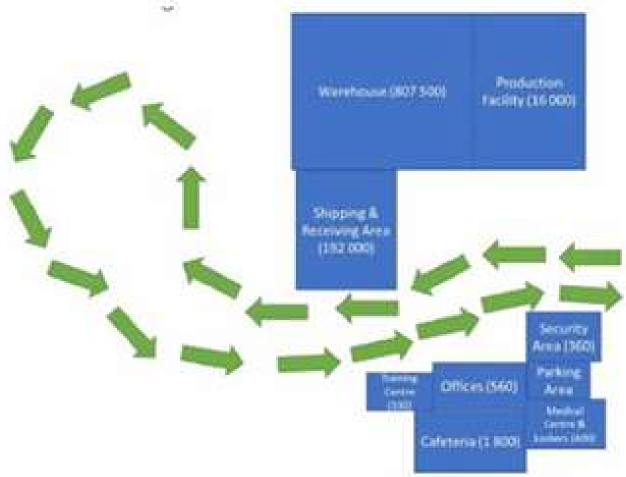
- This is the phase in the Facility Design process, where you really need your creative thinking skills!
- It is important to understand that for facilities, there is no one, right answer.
- As a result, the more time you spend on the creative, brainstorming processes around developing alternative layouts, the more likely you are to come up with a successful design.

Key points of this chapter have been highlighted in these slides but to get the real essence of developing layout alternatives, I would strongly recommend reading through the chapter at least once.

# Block layouts




See how useful these can be when comparing different layout alternatives. We usually do this before we develop detail internal layouts.

# Evaluating alternatives *(Ch 11)*

- Remember at start…*design requirements*?
- Now is the time to **measure against those** – WHY?
  - so that Evaluation is not a Guess….
  - but a Decision based on Facts
  - and Systematically done
- Someone else must be able to also see why and how decisions were made (this is tricky if you are biased)

  - Involve variety of people to help judge
  - Don't be afraid to combine ideas also
  - If alternative ideas are not included in the comparisons, then it cannot (and will not) be selected!
  - Try and focus on data driven solutions and not management ego's or feelings

# Evaluation tools/techniques

- Start with Requirements e.g. was it to…
  - Minimize distance traveled…***so did it***? show it
  - Meet demand…………………..............ditto
  - Maximize space utilization…..............ditto
  - Increase machine utilization…………ditto
  - Decrease material handling costs…..ditto
  - Decrease material movement time….ditto

- Pros/Cons or Advantages/Disadvantages (Fig 11.1&2)
- Ranking (Fig 11.3 illustrates example)
- Weighted factor comparison (Fig 11.4 and Ex 11.1) (quantifying subjective factors are tricky)
- (or if you wish Analytical Hierarch Process AHP)
- Economic evaluations

# The Master Facility Layout *(Ch 12)*

- Remember that you could have had Alternatives for
  only certain areas, aspects or material handling systems

- Intention of the MASTER PLAN (MFP) = the BIG picture
  - The overall ‘picture’…
    - must be **visual** and must be **clear**
  - 2D or 3D (if it is an important factor)
  - (also called) Total Site/Plant layout or Plot Plan
  - Must indicate setting/context of public roads/waterways

- Factors to consider include:
  - Aesthetics (attractiveness, within a setting)
  - Expansion (10 yrs into future)
  - Flow patterns (outside integration to inside)
  - Sustainable energy usage (Smart/Green)

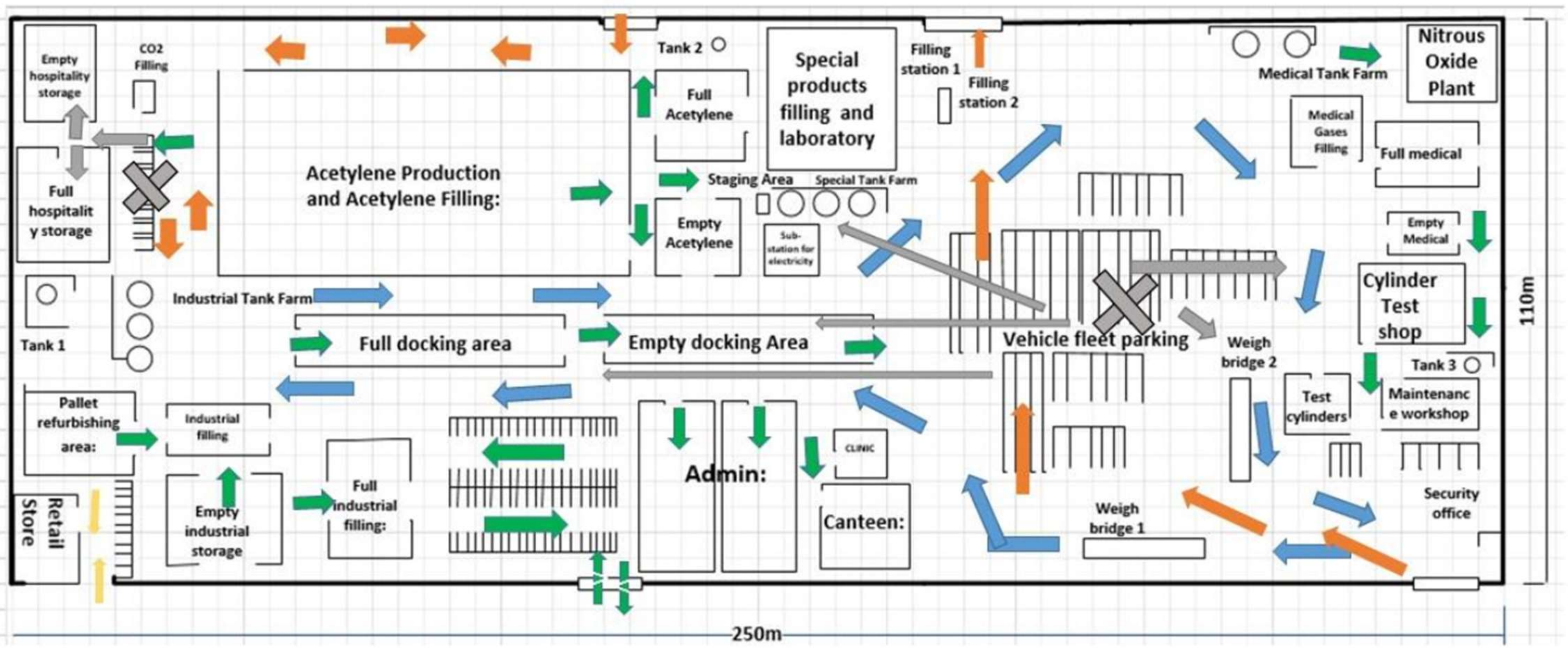
# The Master Facility Layout *(Ch 12)*

- Remember the logic you followed…
  - Start with setting/context of plot/land
  - Indicate outside and inside roads where applicable
  - Receiving and Shipping points
  - Warehouses (if outside)
  - Outside parking, security, offices (if outside manufacturing)
  - Outside walls
  - Production department blocks based on based on Space relationships charts and alternatives
  - Aisles

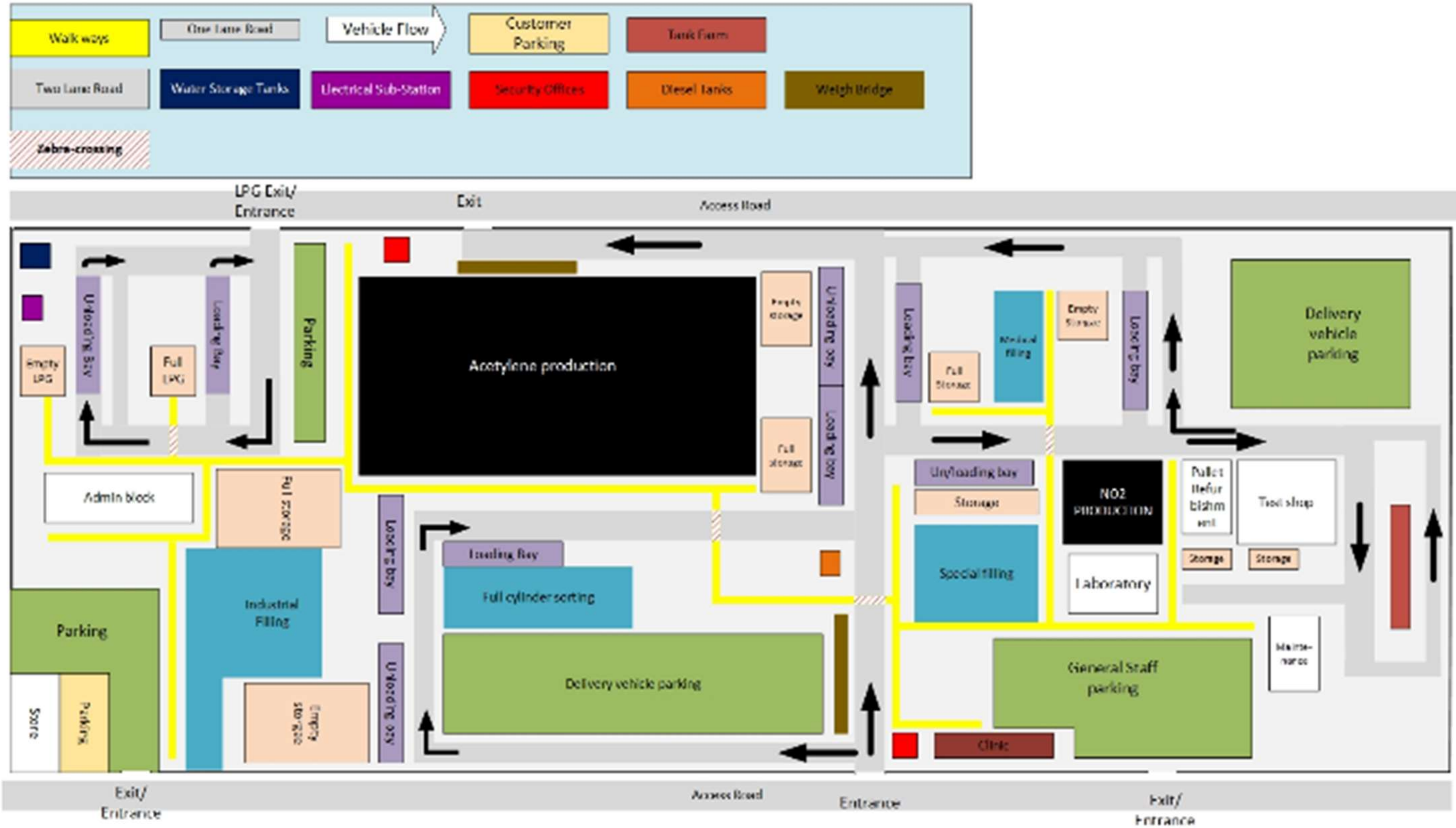
  - Audit the layout….by following material flow
    - Is everything there?
    - Is flow smooth?

  - Does the MFP align with block layouts?

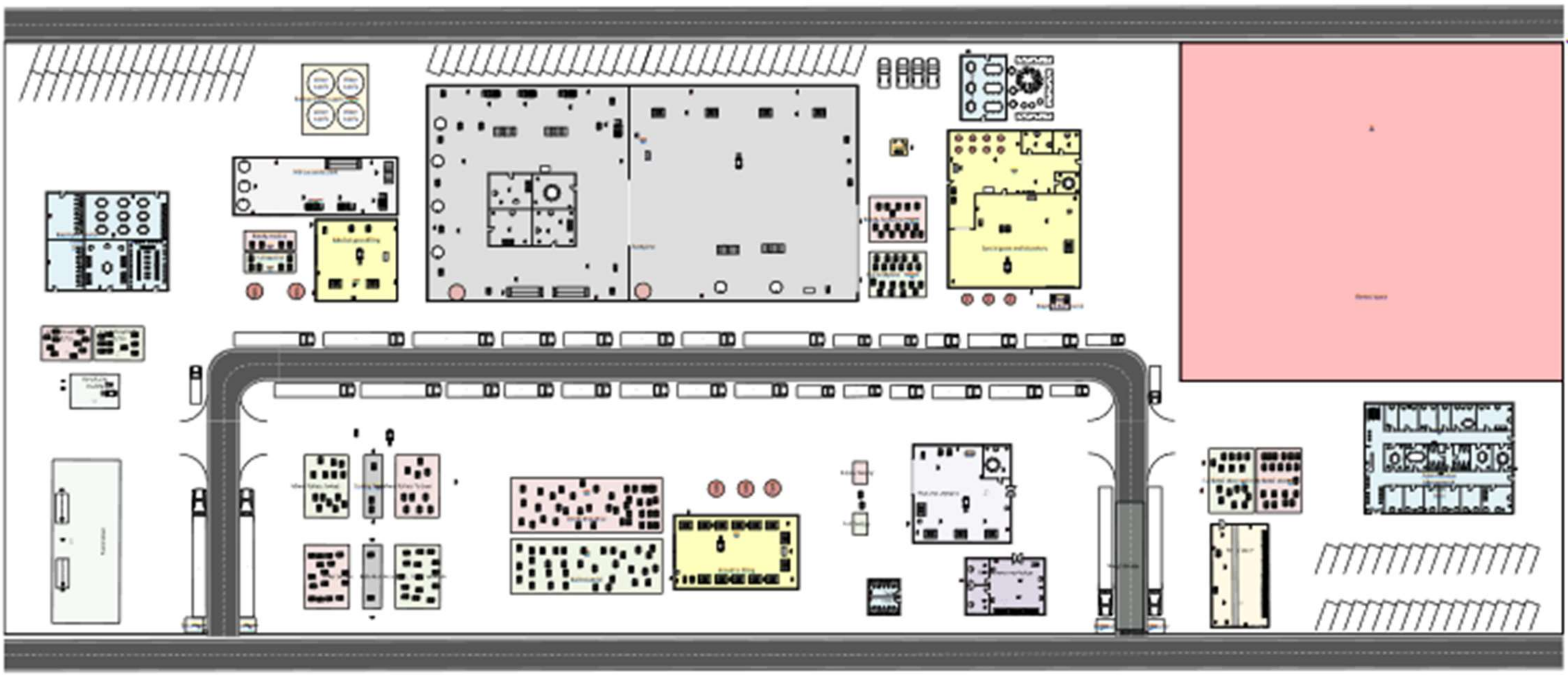
# Some examples *(Ch 12)*



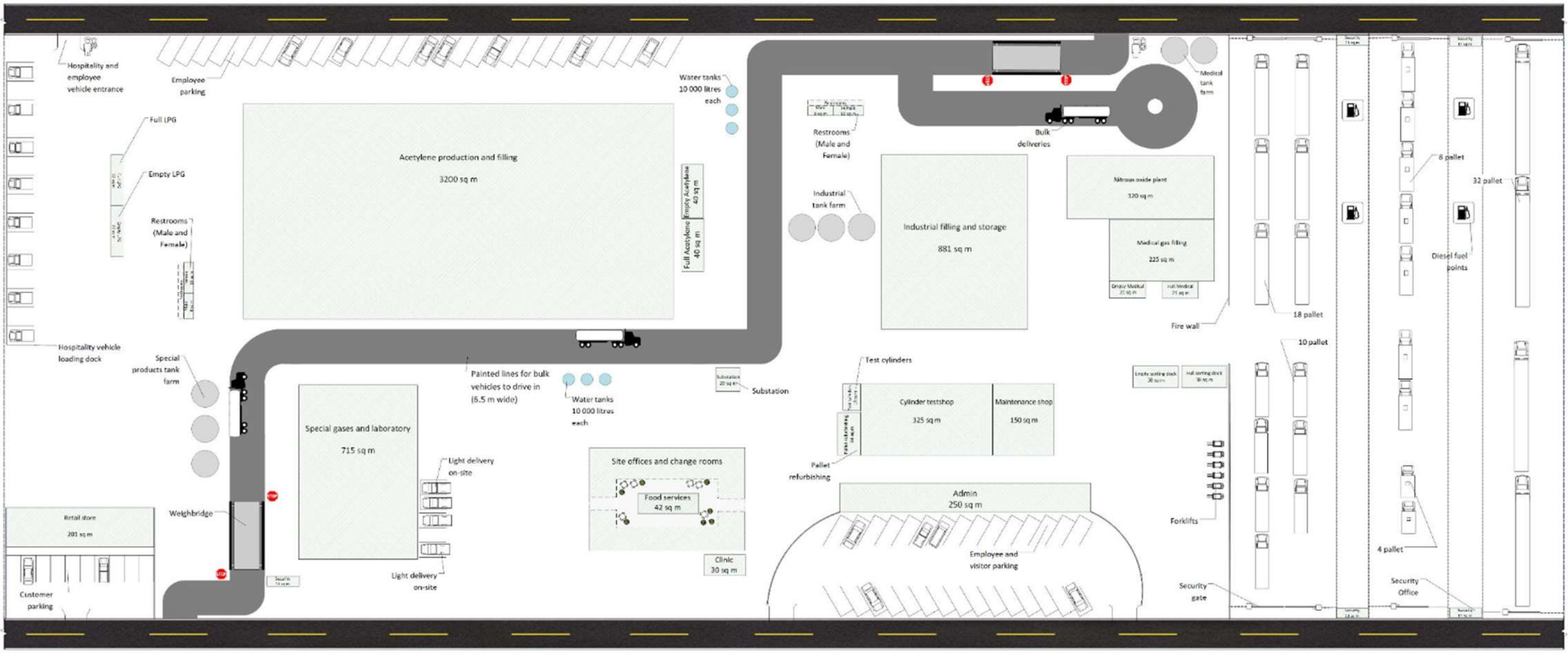

Chanel Bisset, 2019



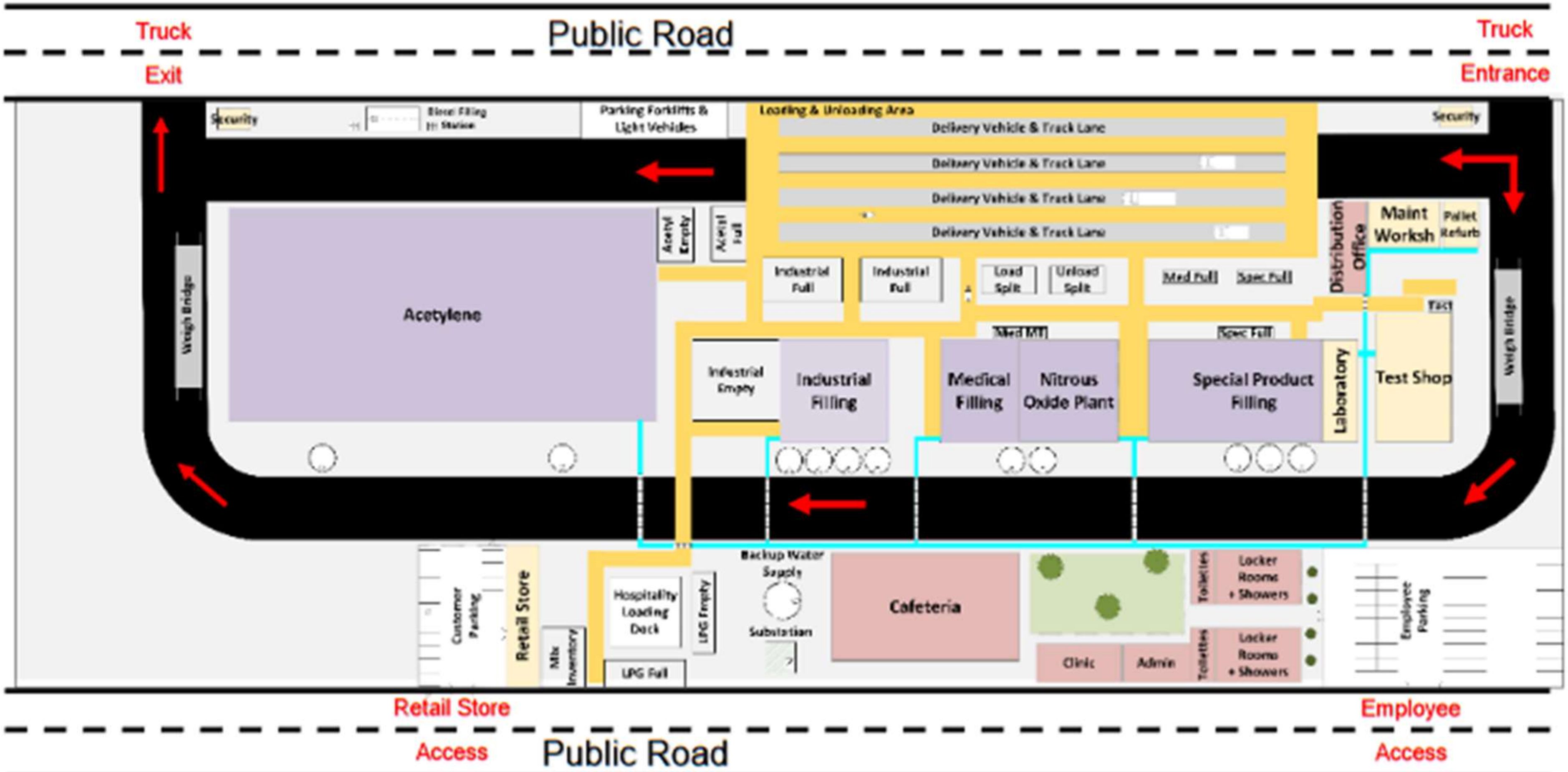

Theuns Bonnet, 2019



Lean Burger, 2019




Marlene Eddie, 2019




Inge Guess, 2019

# The Business case

- Define a ‘case’ for the project = to get the facility there so operations can start
- The start AND ending of the project should have been defined at the start
- Assume that it does not include stocking and starting production (unless specified in the real world)
- Include 3 things:  BUDGET, SCHEDULE and RISKS (for purposes of this subject)
- Budget = includes the CAPITAL expenses for the building of the facility (not the operating expenses)
- Schedule = implementation plan of the building project and purchasing of the equipment
- Risks = things that can go wrong, and mitigation options

# Selling the plan!

- You have WORDS and
  PICTURES to communicate with

  - Words in a Report (refer to guidance) somebody can READ or HEARD in a presentation (if required)

- Often ONLY the MFP is required (but must be able to ***justify*** and ***support*** calculations and decisions)

- For this module – challenge is INTERVIEW = questions asked on what/how/why to authenticate and evaluate practical insight

- Most NB – show your logic, logically 😎